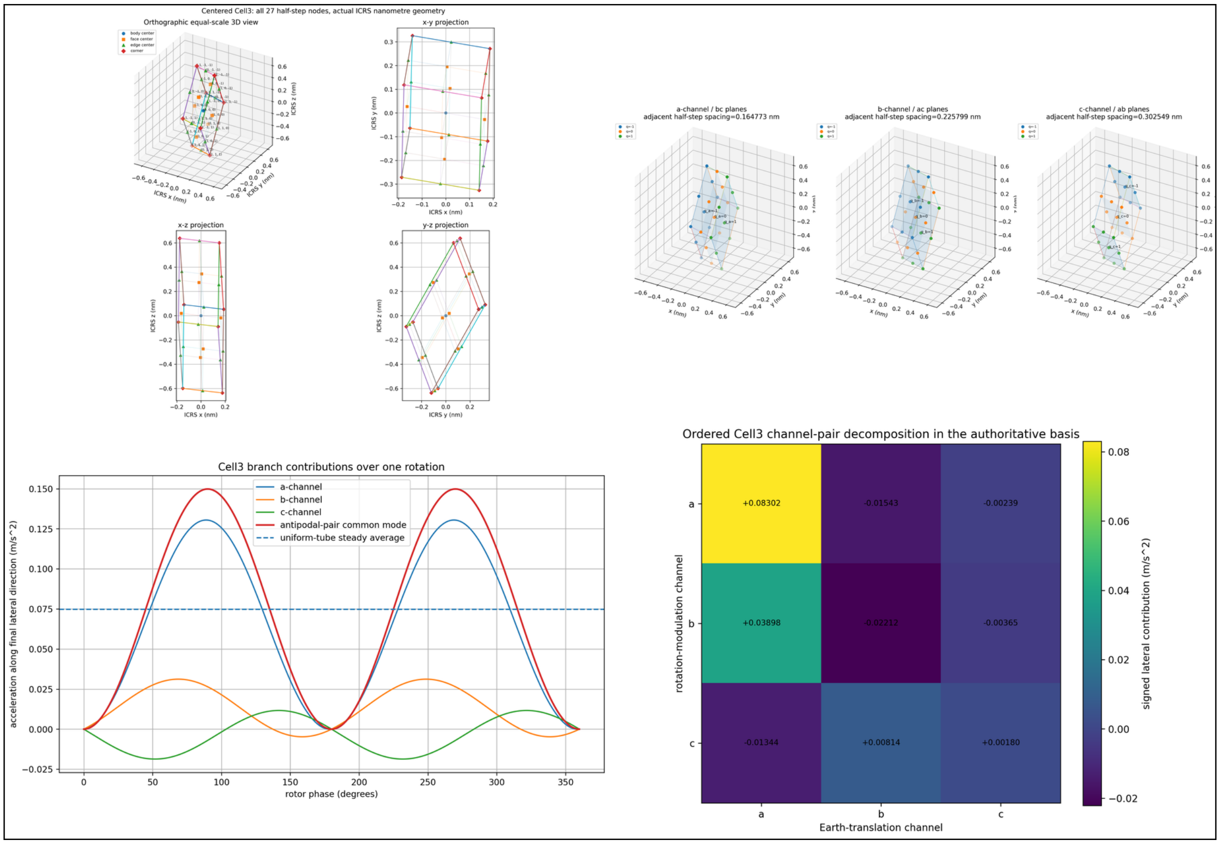
K-FIELD

Single-Cell3 Lateral Force Reduction

Audited Cell3 coordinate reduction, true-scale geometry, and phase-ordered recovery of tubular-rotor lateral acceleration



Title figure. Every panel is regenerated from the physical nanometre Cell3 source kernel, the calibrated Earth-CMB velocity state, and exact calculations. Image aspect ratios are preserved. Cell and plane views use equal physical coordinate scales.

Principal Investigator

P. Dwayne Esterline

BS, Huntington University (1999)

MBA, Indiana Wesleyan University (2008)

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This corrected edition is a standalone educational and archival reference. Every numerical table and figure is regenerated from the source files listed in the archival ledger. The report distinguishes source-defined data, exact algebraic consequences, coordinate-differential interpretations, and the remaining physical localization closure.

Abstract

A rotating tubular mass translating through the fixed K-Field frame follows a translated circular path whose local phase is periodic while its absolute lattice position is open. The physical velocity is mapped into the centered Cell3 half-step address q=2 B^(-1)x. The exact source identity G_phase=4 B^(-T) W B^(-1) converts the translation-rotation scalar v_R^T G_phase V_E into qdot_R^T W qdot_E. This yields the coordinate-differential acceleration a=-(kappa_F/rho)u(qdot_R^T W qdot_E) at material radius rho. The cycle integral is a geometric pump: signed rotational address modulation closes scalarly but not after weighting by rotor radial direction. The result is Delta v_rev=pi kappa_F n x (G_phase V_E), giving 0.0749113841 m/s^2 and 1.69445559 mN for the defined tube. The audit corrects three important interpretations. Source face spacings are primitive boundary spacings; adjacent centered half-step planes are separated by half those values. The bounded q_R variation is an address modulation, not a population of actual reverse face crossings; the total crossing rates remain one-signed. The inertia-equivalent radius is a mechanical surrogate, while cell traversal is resolved across the actual 4-5 mm wall. The exact reduction determines the minimum coordinate kernel. A rule that localizes its impulse exclusively at nodes or plane crossings remains a separate physical closure.

Central result

```
q = 2 B^(-1) x
qdot_E = 2 B^(-1) V_E
qdot_R = 2 B^(-1) v_R
epsilon_cross = kappa_F qdot_R^T W qdot_E
a_cross = -(kappa_F/rho) u (qdot_R^T W qdot_E)
d v = -(kappa_F/rho) u [(W qdot_E)^T d q_R]
Delta v_rev = pi kappa_F n x (G_phase V_E)
```

The exact cell-level object is a phase-correlated differential address-modulation pump. A discrete material carrier must reproduce this kernel without changing its mass, spin, radius, reversal, direction, and cycle invariants.

Contents

1. Purpose, correction boundary, and terminology

The objective is to identify the single-Cell3 coordinate interactions that reconstruct the lateral acceleration of the rotating tube, while preserving the boundary between exact reduction and an unclosed microscopic localization rule. The starting point is the source continuum law. No channel coefficient is fitted to the final force. The physical velocity is transformed into the source Cell3 address coordinates and decomposed there.

The phrase single-Cell3 reduction means that one repeated Cell3 kernel is sufficient to write the interaction everywhere in the translated lattice. The experiment traverses very many cells during one rotation; it is not confined inside one physical cell.

Class	Meaning
Source-defined	Physical Cell3 basis B, direct metric, reciprocal metric, response matrix G_phase, address coupling W, Earth-CMB velocity V_E, calibrated coupling kappa_F, tube geometry, and node/plane registries.
Exact derived	Coordinate transformation, address rates, mixed-energy identity, differential modulation kernel, branch and pairwise decompositions, cycle impulse, radial cancellation, and all numerical evaluations.
Physical closure remaining	The microscopic material state and any rule that localizes the coordinate-differential impulse at a node, face crossing, sheet transition, or distributed intra-cell process.

Corrections applied to the superseded edition

Item	Corrected statement
Cell3 geometry rendering	All 3D Cell3 and plane figures use orthographic projection, one common physical scale on x/y/z, and equal-axis limits. Orthographic x-y, x-z, and y-z views are included to remove perspective ambiguity.
Face-plane spacing	The source face-plane spacing is the full primitive boundary-to-boundary spacing. Adjacent planes of the centered half-step registry are separated by exactly one half of that value. Both values are now printed and plotted.
Absolute trajectory figure	The earlier mixed-unit 3D plot visually enlarged millimetre transverse motion relative to hundreds of metres of translation. The corrected report separates axial advance, true local cross-section, and an explicitly magnified diagnostic path. No combined view is presented as metric aspect.
Trajectory classification	Because V_E is not parallel to the optimal spin axis, the absolute material path is a translated circular or generalized screw/trochoidal path, not a simple circular helix about the spin axis.
Differential event terminology	Positive and negative variation of q_R is an exact differential address-modulation measure. It is not a count of actual reverse face crossings. The total address rates remain one-signed in all three channels for the defined experiment.
Inertia-equivalent radius	R_I is retained only as an exact inertia/energy surrogate. Cell traversal and radial-invariance calculations are resolved across the actual 4-5 mm tube wall.
Discrete localization boundary	The source data establish an exact coordinate-differential kernel. Localizing the impulse exclusively at nodes or plane crossings requires an additional physical interaction rule and is not asserted as already closed.
Pairwise channel interpretation	The 3 x 3 W_ij ledger is an ordered decomposition in the authoritative Cell3 address basis. Individual entries are basis components whose nonlateral parts cancel; they are not independent invariant forces.
Numerical rendering	All vector and scalar tables now contain plain numeric values rather than Python object representations such as np.float64(...).

2. Source data and numerical contract

The physical gravity-state Cell3 kernel is expressed in ICRS Cartesian coordinates with length in nanometres. The ordered basis vectors are the columns of B. Centered half-step coordinates satisfy x=(1/2)Bq. W is defined in the half-step address coordinates. G_phase is dimensionless and closes exactly through the source identity.

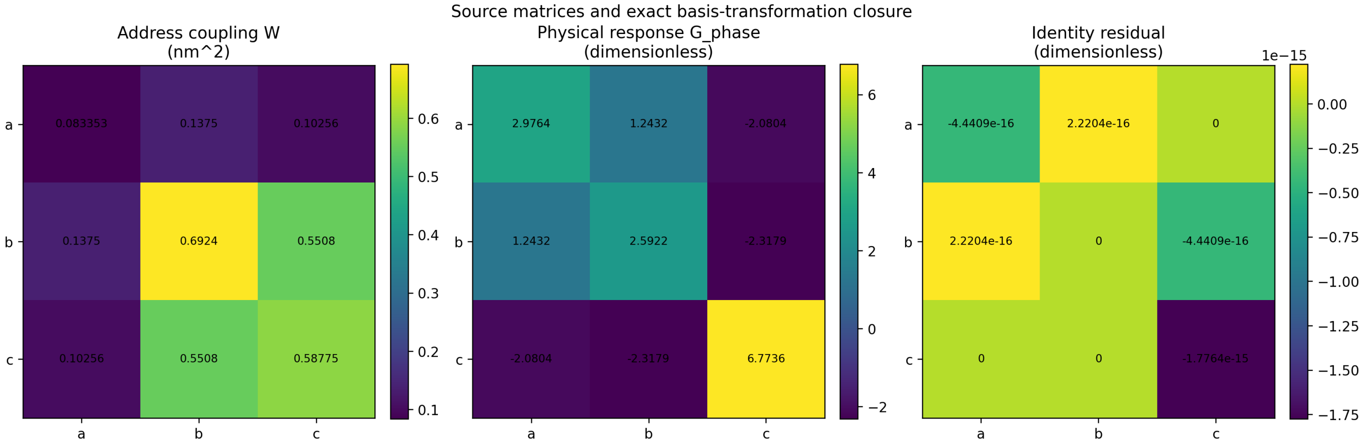

Figure 1. Source matrices and numerical closure. The right panel is G_rebuilt-G_phase and is at machine precision.

Quantity	Definition or value	Dimension
B	Ordered ICRS basis; printed in Appendix A	nm
q	Continuous centered half-step address; integer q selects a centered node	dimensionless
x=(1/2)Bq	Physical location associated with q	nm
W	Address-space coupling matrix	nm^2
G_phase	Physical anisotropic response matrix	dimensionless
G_phase=4B^(-T)WB^(-1)	Direct/address closure identity	dimensionless

Check	Maximum absolute error
G_phase identity	1.776357e-15
Direct metric B^T B	0 nm^2
Reciprocal metric M^(-1)	0 nm^-2
Cell volume	0 nm^3
All 27 source nodes	0 nm
Face areas	0 nm^2
Primitive face spacings	5.551115e-17 nm

3. Dimensionally correct Cell3 geometry

The centered Lucas window contains 27 states q=(i,j,k), with i,j,k in {-1,0,+1}. The corresponding physical location is x=(1/2)Bq. The set contains eight corners, twelve edge centers, six face centers, and one body center. The geometry is oblique and strongly extended in ICRS z; this is a physical property of the source basis rather than a drawing artifact.

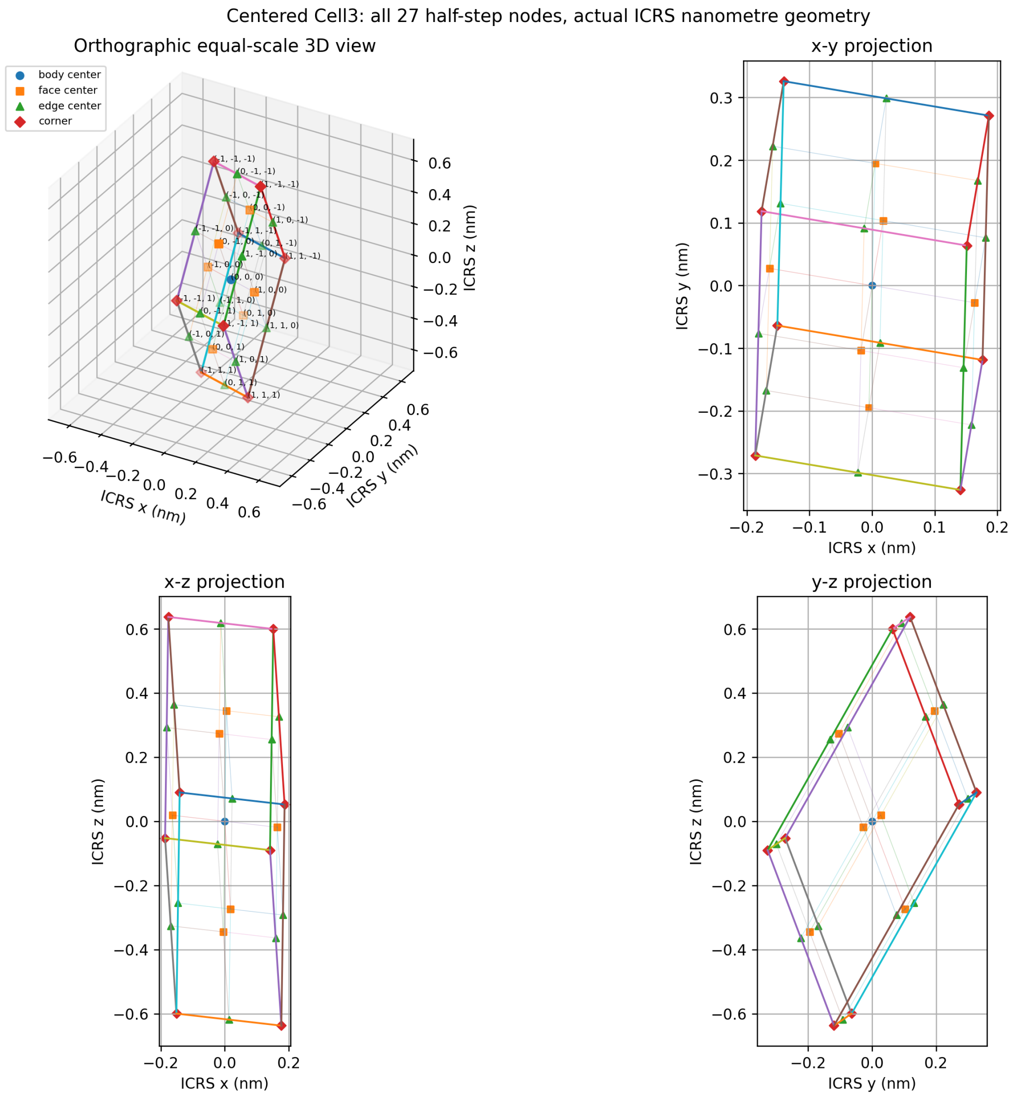

Figure 2. Corrected Cell3 geometry. The 3D panel is orthographic and uses one common physical scale on x, y, and z. The three orthographic projections use equal aspect. All 27 half-step nodes and all nearest-neighbour half-step links are generated from x=(1/2)Bq.

Quantity	Value
|a|	0.334179679073151 nm
|b|	0.586410890414836 nm
|c|	0.792319765037693 nm
angle(a,b)	83.933483012313 deg
angle(a,c)	80.448130018338 deg
angle(b,c)	50.363231813756 deg
Cell volume	0.117914891916699 nm^3

4. Primitive and centered half-step plane families

The reciprocal plane family associated with q_a is the bc family; q_b uses ac; q_c uses ab. The source interplane spacing is the distance between the two primitive cell boundary planes. Because x=(1/2)Bq, adjacent integer q planes in the centered half-step registry are separated by exactly one half of the primitive spacing. The centered window therefore contains q_i=-1,0,+1 planes.

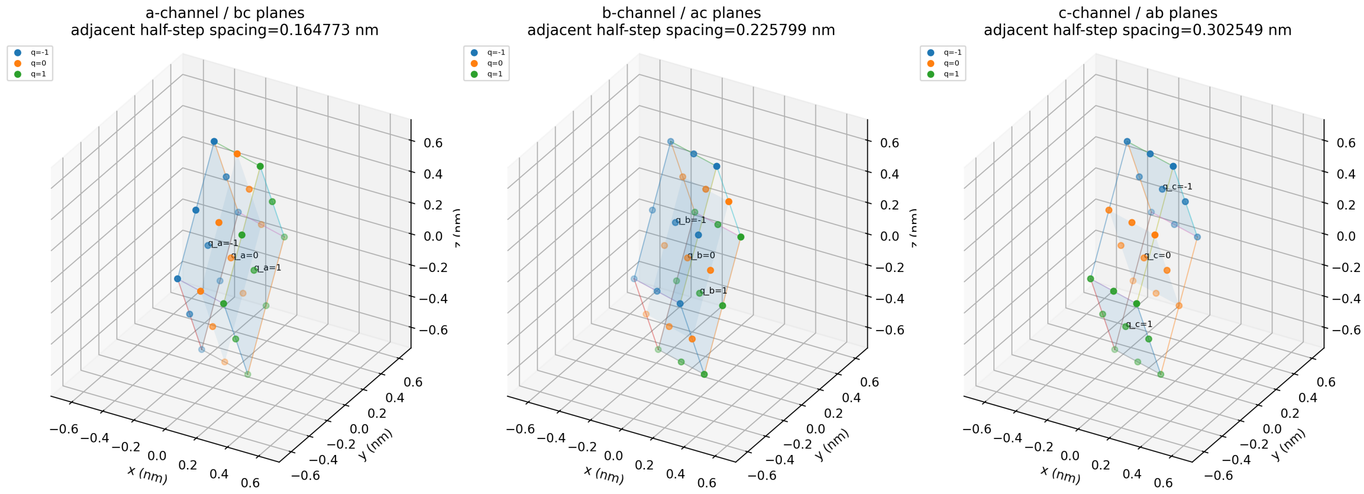

Figure 3. Corrected plane registry. Each panel is orthographic, uses equal x/y/z physical scale, displays all 27 nodes, and shows the three q_i=-1,0,+1 half-step planes. The displayed spacing is the adjacent half-step spacing.

Channel	Physical face family	Area (nm^2)	Primitive boundary spacing (nm)	Adjacent half-step spacing (nm)	Unit normal (ICRS)
a	bc	0.357809539129	0.329546529709	0.164773264855	(-0.9959651903, 0.0836021625, -0.0326192906)
b	ac	0.2611062358	0.451597379724	0.225798689862	(0.0890525941, 0.8665767693, -0.4910339483)
c	ab	0.19486916547	0.605097741515	0.302548870757	(0.1940922008, 0.9131771688, 0.3583792348)

5. Tube geometry, mass, spin inertia, and radius distribution

The tube has 10 mm outside diameter, 1 mm wall thickness, 8 mm inside diameter, 100 mm length, and material density 8000 kg/m^3. Its mass is integrated over the actual annulus. Every material radius between 4 and 5 mm is retained in the cell-traversal calculation.

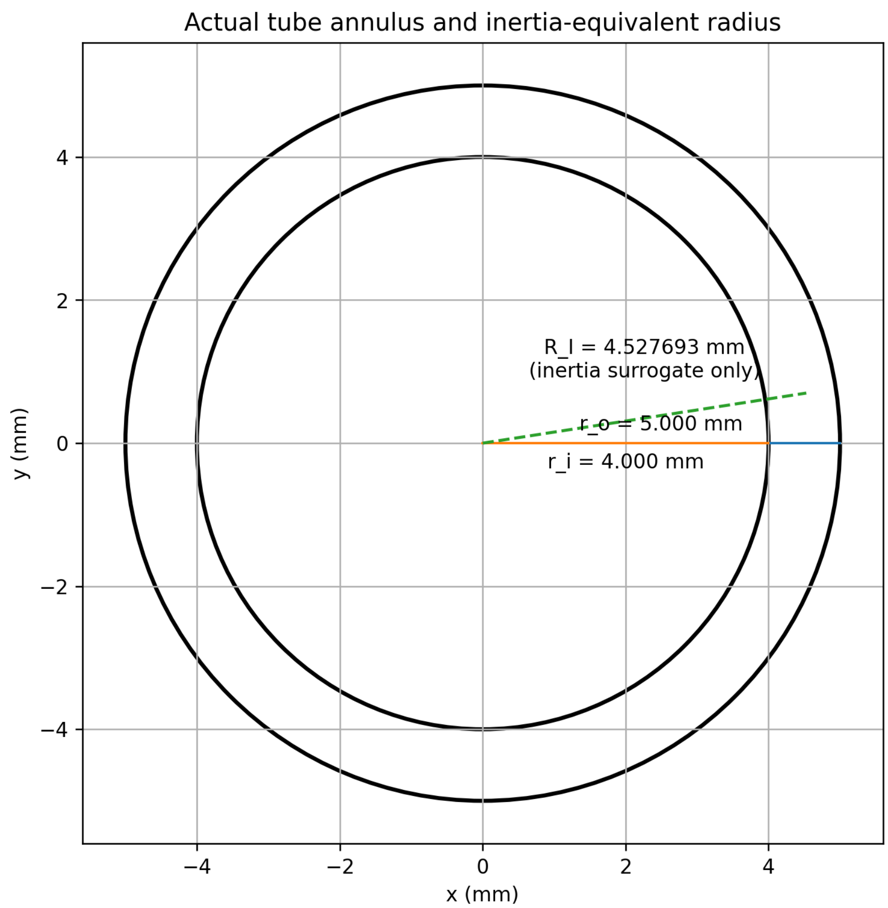

Figure 4. True-scale tube cross-section. R_I is the radius of an equal antipodal point pair with the same axial inertia; it is not used as the unique physical path radius in the corrected cell-level calculation.

Quantity	Value	Unit
Material density	8000	kg/m^3
Outer radius	0.005	m
Inner radius	0.004	m
Length	0.1	m
Metal cross-sectional area	2.82743338823081e-05	m^2
Volume	2.82743338823081e-06	m^3
Mass	0.02261946710585	kg
Axial moment of inertia	4.63699075669854e-07	kg m^2
Inertia-equivalent radius R_I	0.00452769256907	m
Spin speed	50000	rpm
Angular speed	5235.987755982988	rad/s
Spin period	0.0012	s
Stored axial rotational energy	6.356286719461	J
Axial angular momentum	0.002427922683	kg m^2/s

6. Earth-CMB state and maximum pure-lateral orientation

The fixed external state is V_E. The anisotropic response vector is g=G_phase V_E. To maximize force while preserving exact orthogonality to the Earth-CMB axis, the spin axis is the normalized projection of the Earth-CMB unit vector into the plane perpendicular to g. The resulting lateral direction is n_opt x g.

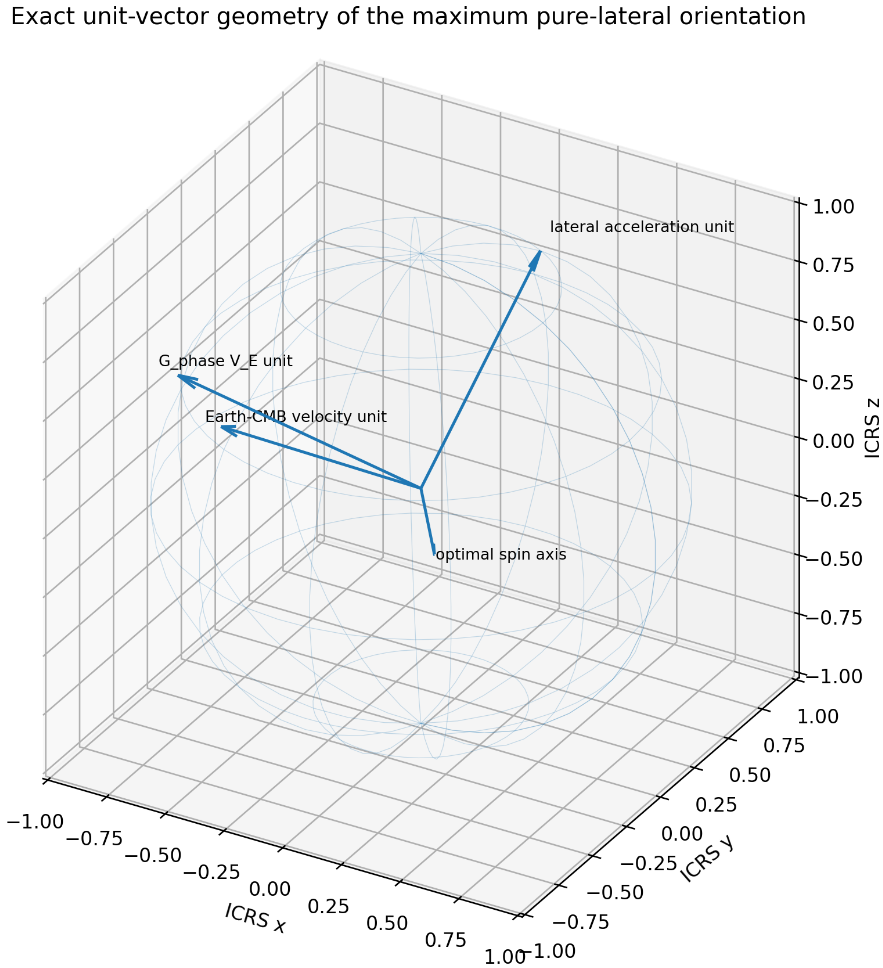

Figure 5. Equal-scale orthographic unit-sphere geometry. Every arrow is a unit vector computed from the numerical ledger.

Vector	ICRS components	RA	Dec
V_E unit	(-0.970770855836, 0.207348009111, -0.120874929482)	167.943300 deg	-6.942600 deg
G_phase V_E unit	(-0.945636157362, -0.15460629331, 0.28612785946)	189.285391 deg	16.626278 deg
Optimal spin axis	(-0.315865036407, 0.646162683024, -0.694768354089)	116.050867 deg	-44.008761 deg
Lateral direction	(0.077469585422, 0.747375863362, 0.65986951907)	84.082114 deg	41.289922 deg

```
n_opt = [c_hat - (c_hat dot g_hat) g_hat] / ||c_hat - (c_hat dot g_hat) g_hat||
a_hat = (n_opt x g) / ||n_opt x g||
a_hat dot c_hat = 0
```

7. Absolute translated-circle trajectory through the fixed lattice

For a material radius rho and spin sense sigma=+1 or -1, the absolute trajectory is r_sigma(t)=r_0+V_E t+rho[cos(omega t)e1+sigma sin(omega t)e2]. Reversing the motor reverses local chirality but does not reverse Earth translation. Because V_E is not parallel to n_opt, the absolute path is a translated circular or generalized screw/trochoidal path rather than a simple circular helix about the spin axis.

```
r_sigma(t) = r_0 + V_E t + rho[cos(omega t)e1 + sigma sin(omega t)e2]
r_plus(t) - r_minus(t) = 2 rho sin(omega t)e2
full time reversal additionally requires V_E -> -V_E
```

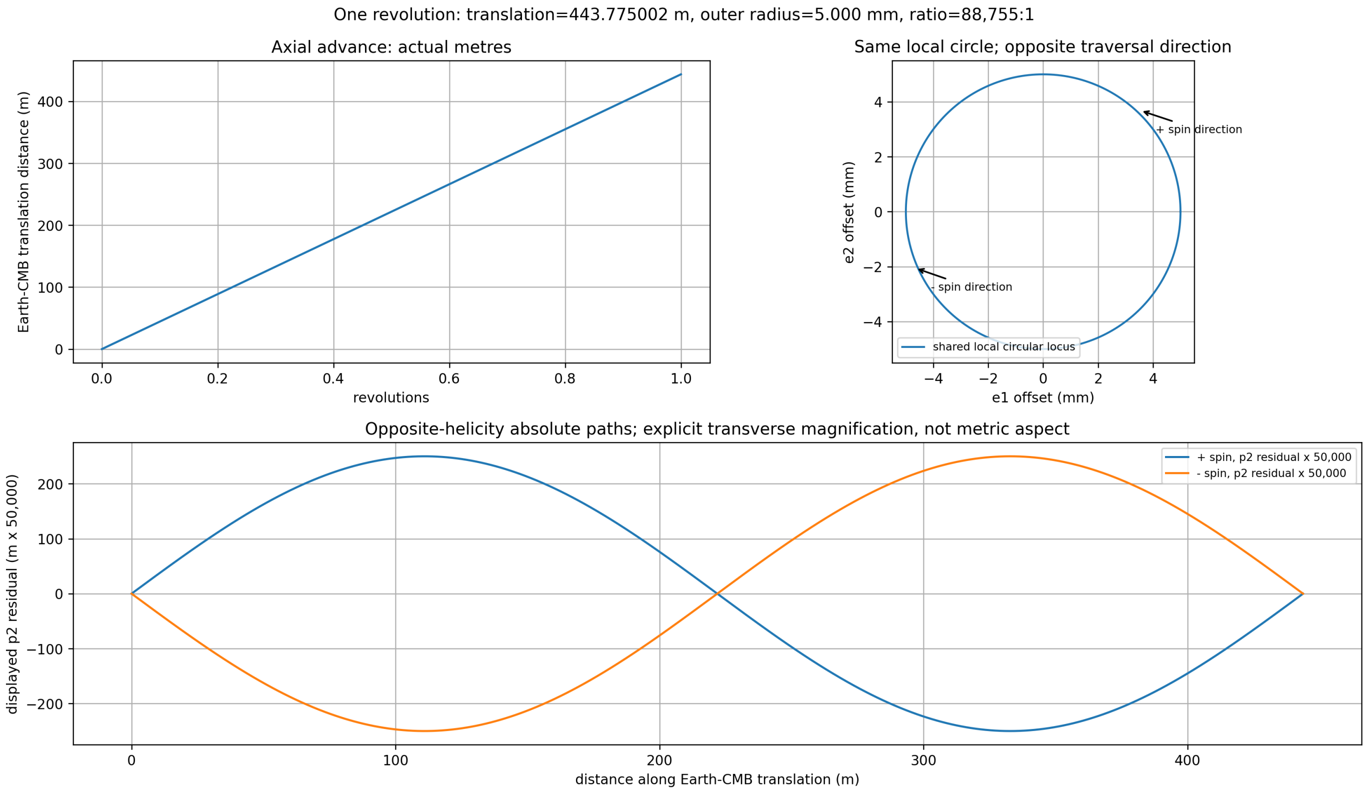

Figure 6. Correct dimensional views of one outer-surface revolution. Axial translation and local millimetre motion are shown on separate physical axes. The lower diagnostic path explicitly multiplies the transverse residual by 50,000 and is not presented as metric aspect.

Quantity	Value
Earth-CMB translation per revolution	443.775002400488 m
Outer-radius to translation ratio	1 : 88,755.000
Translation component along spin axis	232.801345666941 m/rev
Translation component in rotor plane	377.809192862236 m/rev
Angle between V_E and spin axis	58.359138729 deg

8. Continuum force law and exact address transformation

For one material element at radius rho, the source translation-rotation term is radial. Circumferential integration cancels the even self-motion resultant and retains the term odd in signed angular speed. The exact half-step coordinate transformation converts the same law into Cell3 address rates.

```
a_cross(theta,rho) = -kappa_F omega [t(theta)^T G_phase V_E] u(theta)
x = (1/2)Bq
qdot = 2 B^(-1) v
qdot_E = 2 B^(-1) V_E
qdot_R = 2 B^(-1) v_R
v_R^T G_phase V_E = qdot_R^T W qdot_E
```

The transformation is dimensionally exact: B^(-1) has units 1/m, velocity has m/s, qdot has 1/s, and qdot_R^T W qdot_E has m^2/s^2.

Result	Value
Mean acceleration vector	(0.00580335386756, 0.05598696034797, 0.04943173898179) m/s^2
Mean acceleration magnitude	0.07491138407399 m/s^2
Tube force vector	(0.00013126877191, 0.00126639520795, 0.00111811959388) N
Tube force magnitude	0.00169445558792 N
Velocity increment per revolution	(6.96402464107689e-06, 6.71843524175683e-05, 5.93180867781433e-05) m/s
Increment magnitude	8.98936608887906e-05 m/s

9. Actual monotonic crossing rates and rotational timing modulation

The physical trajectory uses q_total=q_E+q_R. For the defined tube, the Earth-driven rates exceed the rotational modulation in every branch. Therefore the actual total crossing rate never changes sign. Rotation advances and retards the timing of an enormous monotonic sequence; it does not create equal populations of forward and reverse physical plane crossings.

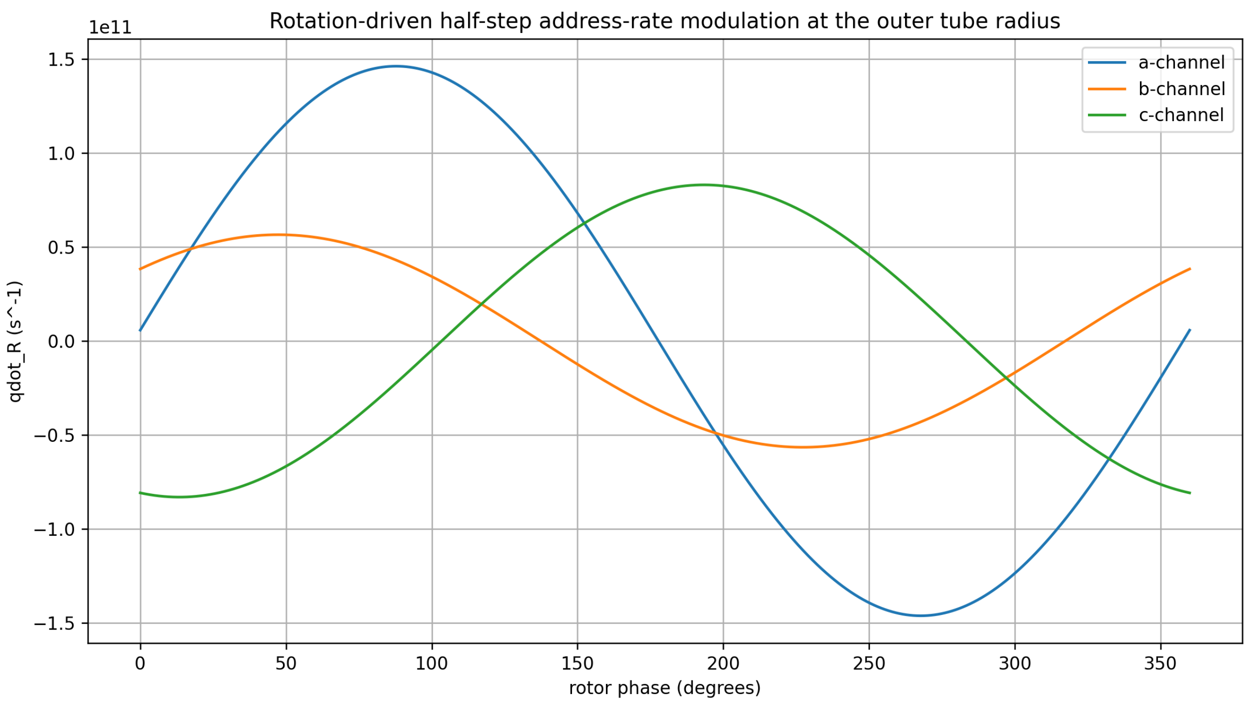

Figure 7. Rotation-driven address-rate modulation at the actual outer tube radius.

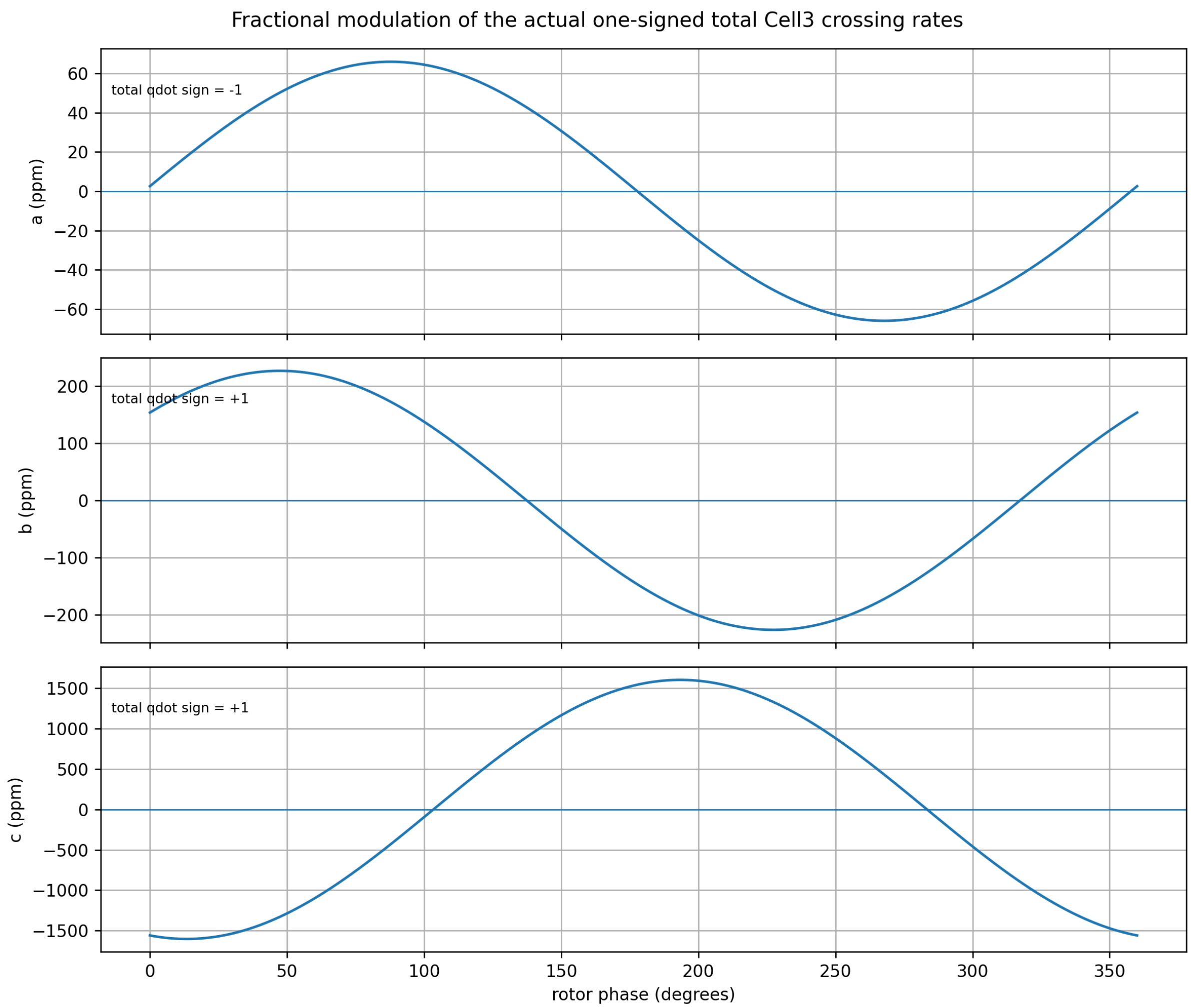

Figure 8. Fractional modulation of the total rates in parts per million. The total qdot sign remains fixed in all three branches.

Channel	qdot_E (s^-1)	qdot_R amplitude at 5 mm (s^-1)	Fractional modulation	Total qdot range (s^-1)	Actual sign
a	-2.217734842862e+15	1.462174259488e+11	6.593097746532e-05	[-2.217881060e+15, -2.217588625e+15]	-1
b	2.499065863112e+14	5.655311070343e+10	0.000226297	[2.498500332e+14, 2.499631394e+14]	+1
c	5.181751775713e+13	8.307598725781e+10	0.001603241353	[5.173444177e+13, 5.190059374e+13]	+1

Actual crossing-time modulation

If t_i,k^0 is an Earth-only half-step crossing time, the first-order timing shift caused by the bounded rotational address is delta t_i,k = -q_R,i(t_i,k^0)/qdot_E,i. The approximation error is second order in |qdot_R/qdot_E| and is below the printed modulation ratios.

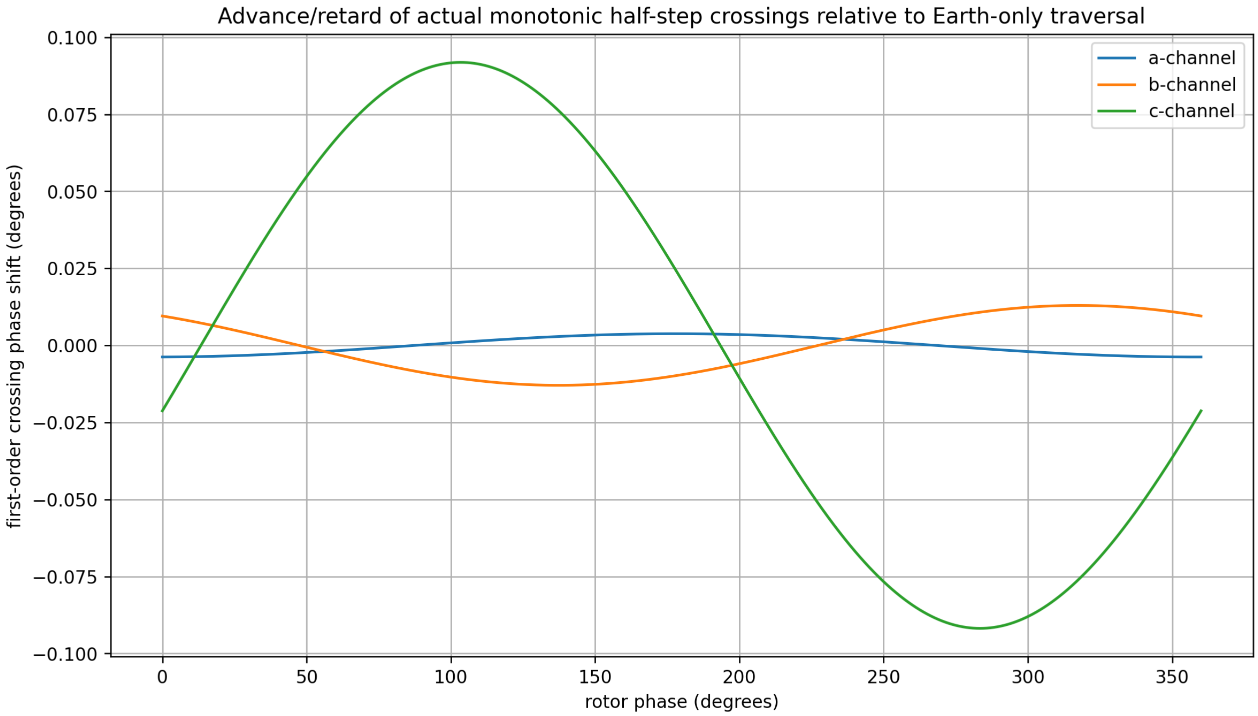

Figure 9. First-order advance/retard of the actual monotonic crossing sequence. The largest envelope is 0.0919 degree in the c-channel.

Channel	Baseline address advance/rev	q_R amplitude (half-step units)	Max time shift (s)	Max rotor-phase shift (deg)
a	2.661281811435e+12	2.792547132711e+07	1.259188915978e-08	0.0037775667
b	2.998879035735e+11	1.080084853881e+07	4.321954334312e-08	0.012965863
c	6.218102130856e+10	1.586634482918e+07	3.061965434846e-07	0.091858963

10. Mixed Cell3 interaction and coordinate-differential kernel

The exact mixed energy per unit mass is epsilon_cross=kappa_F qdot_R^T W qdot_E. Radial differentiation at material radius rho gives the acceleration. Defining h_E=Wqdot_E exposes the three address components.

```
epsilon_cross = kappa_F qdot_R^T W qdot_E
a_cross = -(kappa_F/rho) u (qdot_R^T W qdot_E)
h_E = W qdot_E
d v = -(kappa_F/rho) u [h_E^T d q_R]
```

A unit of signed d q_R in branch i contributes the coordinate-differential velocity increment -(kappa_F/rho)h_E,i u. This statement is an exact line-integral decomposition. It does not, by itself, prove that the physical impulse is concentrated at one discrete face boundary. Actual localization can be node-centered, plane-localized, distributed through the cell, or stateful; the source data do not select among those composition rules.

Correct interpretation of positive and negative modulation

Over one cycle, q_R,i returns to its starting value. Its positive and negative total variations are equal. These numbers quantify how much the rotational path advances and retards the half-step coordinate relative to the Earth-only trajectory. They are not actual reverse-crossing counts because qdot_total never changes sign.

Channel	Positive q_R variation/rev	Negative q_R variation/rev	Continuous baseline crossing measure/rev
a	5.585094265422e+07	5.585094265422e+07	2.661281811435e+12
b	2.160169707762e+07	2.160169707762e+07	2.998879035735e+11
c	3.173268965837e+07	3.173268965837e+07	6.218102130856e+10

A discrete face-crossing representation can be introduced by writing q_i=N_i+phi_i, where N_i is an integer crossing count and phi_i is intra-cell phase. The exact differential kernel then contains both dN_i and dphi_i. Assigning all impulse to dN_i requires an additional localization law.

11. Radius-resolved reduction across the physical tube wall

At material radius rho, qdot_R is proportional to rho, while the coefficient multiplying one unit of d q_R is proportional to 1/rho. Their product is independent of rho. This is the actual reason the leading acceleration is uniform across the wall at fixed angular speed.

```
qdot_R(rho) = 2 B^(-1) [rho omega t]
d v / d q_R = -(kappa_F/rho) u W qdot_E
qdot_R x (d v/d q_R) is independent of rho
```

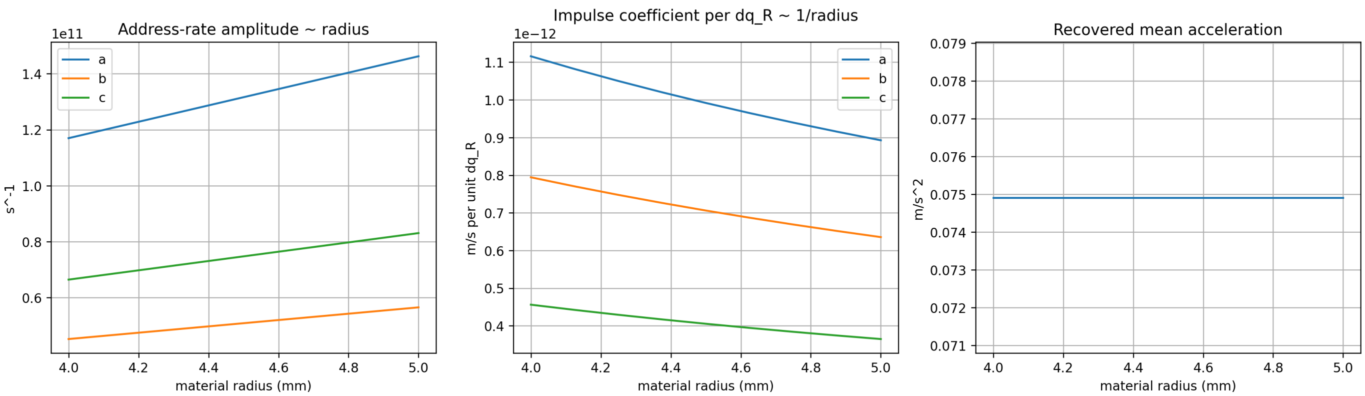

Figure 10. Exact radial cancellation across the actual 4-5 mm material wall. The left and centre panels vary inversely; the recovered mean acceleration is constant to numerical precision.

Radius (mm)	Mean acceleration magnitude (m/s^2)	Error from exact result (m/s^2)
4	0.07491138407399	4.78544781e-17
4.5	0.07491138407399	5.02246636e-17
5	0.07491138407399	5.28521592e-17

The three-point summary shows the inner, middle, and outer wall radii; Appendix F prints the complete nine-radius ledger. The inertia-equivalent radius R_I remains useful for matching axial inertia, spin energy, and angular momentum with an antipodal pair. It is not required for the cell-coordinate force reduction and does not replace the actual radial material distribution.

12. Branch-resolved acceleration and geometric pumping

The address coupling separates the mixed acceleration into a, b, and c rotational modulation branches. A single antipodal pair retains a 2 omega phase ripple; the complete tube occupies all circumferential phases and retains only the steady mean.

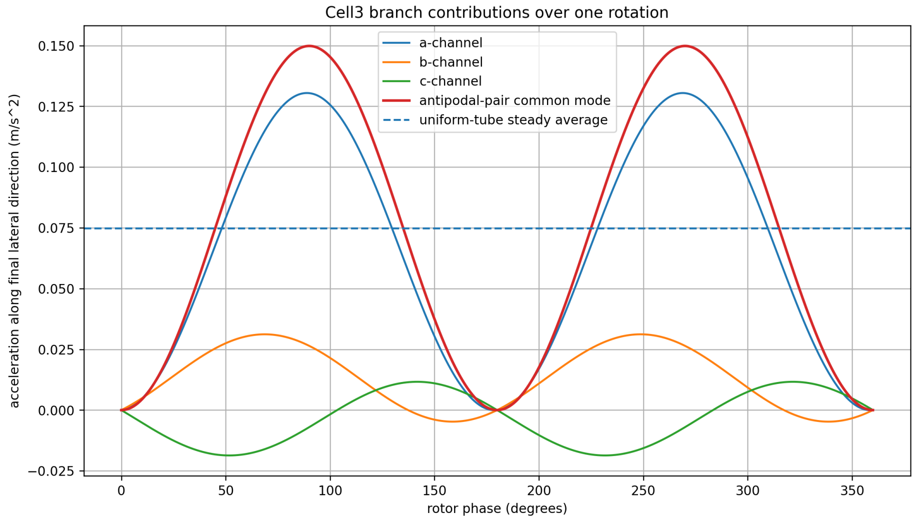

Figure 11. Instantaneous branch contributions along the final lateral direction. The uniform-tube result is the dashed steady average.

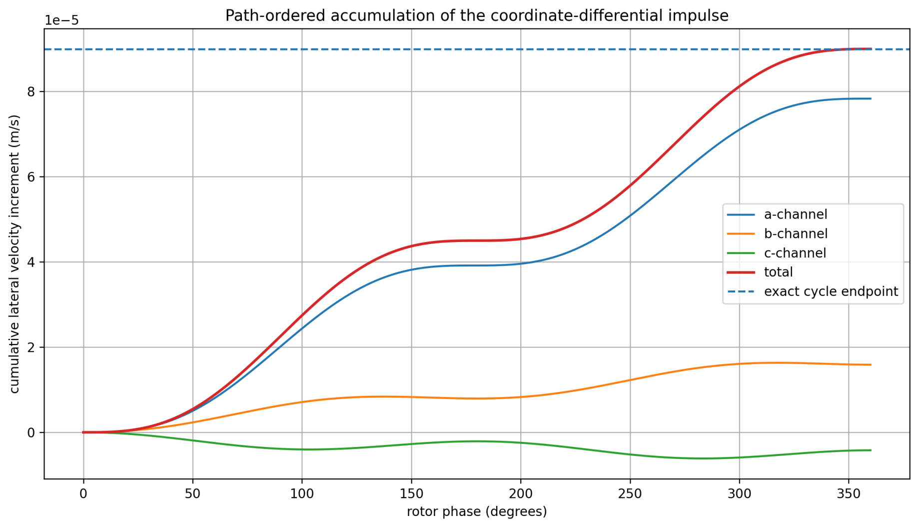

Figure 12. Path-ordered accumulation of the coordinate-differential impulse. The exact endpoint equals pi kappa_F n x (G_phase V_E).

Channel	Mean lateral acceleration (m/s^2)	Share of final output	Cycle Delta v along output (m/s)
a	0.06520446351145	87.042129%	7.82453562137362e-05
b	0.01321422755063	17.639812%	1.58570730607513e-05
c	-0.00350730698808	-4.681941%	-4.20876838569668e-06

Geometric pump identity

```
integral_cycle d q_R = 0
integral_cycle u d q_R^T != 0
Delta v_cycle = -(kappa_F/rho) integral u d q_R^T W qdot_E
Delta v_cycle = -kappa_F integral u d u^T G_phase V_E
integral u d u^T = -pi [n]_x
Delta v_cycle = pi kappa_F n x (G_phase V_E)
```

13. Ordered 3 x 3 channel-pair decomposition

The scalar qdot_R^T W qdot_E contains nine ordered products. The row index identifies the rotation-modulation coordinate; the column identifies the Earth-translation coordinate. Each entry produces a full vector. The matrix below shows only the signed component along the final lateral direction. Individual entries also contain nonlateral components that cancel in the full sum.

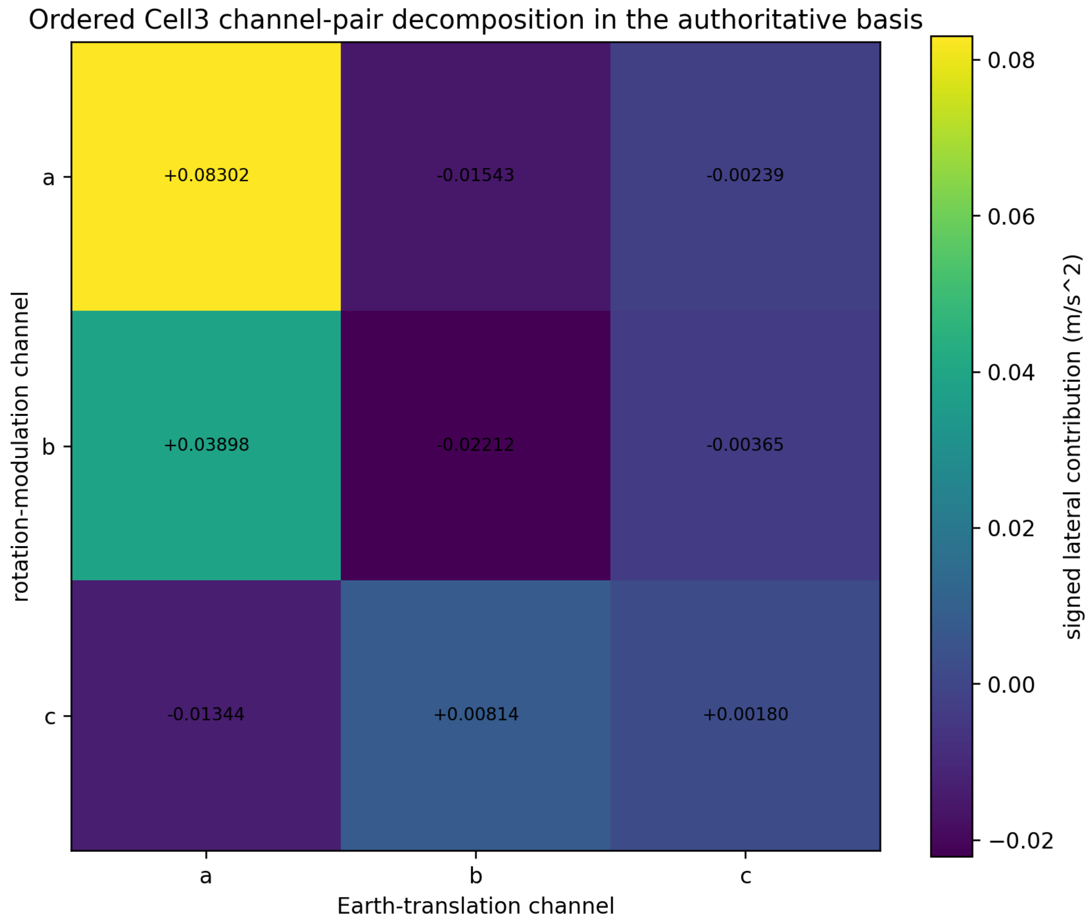

Figure 13. Ordered W_ij decomposition in the authoritative address basis. The entries are coordinate components, not independent invariant forces.

Rotation / Earth	a	b	c
a	+0.083024520468	-0.015433206440	-0.002386850517
b	+0.038984263299	-0.022121266138	-0.003648769610
c	-0.013442370320	+0.008135119008	+0.001799944324

The pairwise vector sum reconstructs the exact continuum acceleration with error 2.046e-16 m/s^2. The dominant positive coordinate term is rotation-a / Earth-a: +0.083024520468 m/s^2. The largest negative coordinate term is rotation-b / Earth-b: -0.022121266138 m/s^2.

14. Recovery of the uniform-tube force

Every material radius has the same mean acceleration at fixed omega. Axial integration contributes only mass, and radial annular integration reproduces the tube mass. The exact result is therefore independent of how the mass is distributed across the 4-5 mm wall, provided the tube is uniform in phase and spins rigidly.

```
a_bar = (kappa_F omega/2) n x (G_phase V_E)
F_tube = M_tube a_bar
Delta v_rev = a_bar (2 pi/omega) = pi kappa_F n x (G_phase V_E)
```

Quantity	Value
Acceleration vector	(0.005803353867564, 0.055986960347974, 0.049431738981786) m/s^2
Acceleration magnitude	0.074911384073992 m/s^2
Force vector	(0.000131268771911, 0.001266395207947, 0.001118119593883) N
Force magnitude	0.001694455587915 N
Cycle velocity increment vector	(6.964024641076892e-06, 6.718435241756831e-05, 5.931808677814329e-05) m/s
Cycle increment magnitude	8.989366088879059e-05 m/s
Numerical cycle endpoint error	1.95486690e-19 m/s
Wall integration error	5.60506309e-17 m/s^2

15. Node-centered field and composition boundary

The source node-centered law defines an anisotropic convergent direction around any selected Cell3 node: g_L(r_L)=-Gamma(rho_L)G_phase r_L/rho_L, with rho_L^2=r_L^T G_phase r_L. It supplies local metric stiffness and convergence.

This local field family is translationally repeated with fixed ICRS orientation. The source leaves the global composition of many node-centered responses as a separate closure. The tube reduction therefore cannot silently replace the mixed translation-rotation kernel with an unspecified sum of static node gradients.

15.1 Node-centered direction geometry

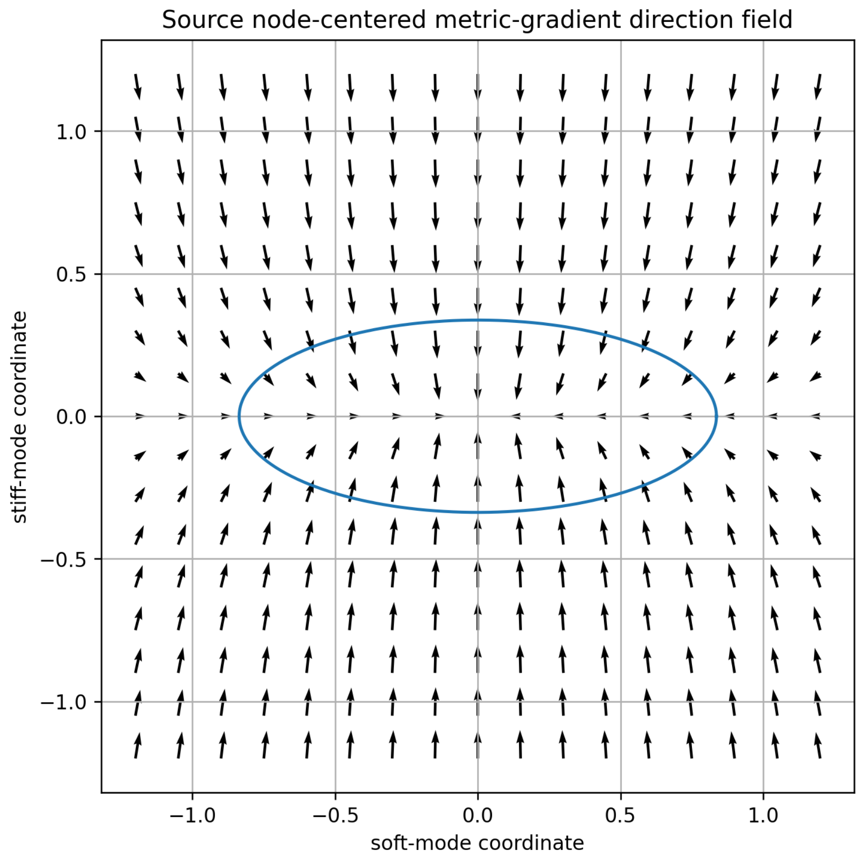

Figure 14. Source node-centered direction field in the plane of the softest and stiffest G_phase modes. The arrows are actual negative metric gradients.

A position-only, memoryless periodic node field cannot by itself select the observed spin-odd tube force, because a uniform rim has the same instantaneous mass occupancy under spin reversal. The leading lateral channel requires the directed translation-rotation flux. The node field may still supply the local material interaction, stiffness, or transition geometry. A global composition rule must specify how the family of node-centered fields acts along the absolute translated path.

16. Polarity, registry, forward/reverse diagnostics, and remaining closure

One integer half-step changes lambda=i+j+k by one and reverses chi=(-1)^lambda. A two-step return restores polarity. The exact differential kernel does not require an added polarity multiplier; its sign is carried by d q_R and the ordered W coupling. Polarity, sheet, and registry become independently observable when the response retains memory between successive cell phases.

```
memoryless kernel: d v = -(kappa_F/rho) u [(W qdot_E)^T d q_R]
stateful extension: z_(j+1) = T(z_j, q_j, lambda_j, chi_j, Delta t_j)
```

Forward and reverse operation follow different absolute translated-circle paths. Define a_odd=(a_+-a_-)/2 and a_even=(a_++a_-)/2. The homogeneous instantaneous kernel predicts a_even=0. A nonzero a_even after apparatus controls would isolate absolute cell phase, intra-cell localization, material relaxation, registry memory, occupancy dependence, or noncommuting sheet order.

Observable	Leading result	Cell-level role
Spin-odd acceleration	Reverses with signed omega	Mixed Earth/rotation address flux
Spin-even residual	Zero in homogeneous memoryless reduction	Diagnostic for path history or additional local composition
2 omega pair ripple	Present for one antipodal pair; removed by full rim	Incomplete versus complete circumferential phase occupancy
Sidereal/seasonal envelope	Follows V_E(t)	Changing lattice traversal vector in fixed ICRS geometry
Crossing direction	One-signed in all three channels here	Rotation changes timing, not crossing polarity

17. Scaling laws, experimental program, and principal findings

Change	Coordinate-level scaling	Macroscopic consequence
Mass M	Velocity increment per unit mass unchanged	Force proportional to mass; free acceleration mass-independent
Angular speed omega	qdot_R proportional to omega	Mean acceleration and force proportional to omega
Spin reversal	d q_R changes sign	Spin-odd force reverses
Material radius rho at fixed omega	qdot_R proportional to rho; coefficient per d q_R proportional to 1/rho	Leading acceleration independent of rho
One revolution	q_R returns to start; vector-weighted integral remains	Delta v_rev independent of omega and rho
No rotation	d q_R=0	No spin-odd lateral channel

Experimental and computational program

Test	Procedure	Discriminating result
Simultaneous counter-rotation	Operate matched tubes at the same location and epoch	Odd component tests the leading kernel; common residual tests spin-even path effects
Start-phase sweep	Repeat at controlled absolute rotor phase	Tests sensitivity to intra-cell phase and discrete localization
Radius sweep	Use equal-mass tubes with different wall radii at fixed omega	Leading acceleration constant; localization corrections may expose cell scale
Angular-rate sweep	Hold geometry fixed and vary omega	Linear leading law; departures locate temporal response or coherence corrections
Sidereal/seasonal run	Track fixed laboratory orientation through V_E(t)	Tests fixed ICRS anisotropy and Q-weighted translation
Material comparison	Match geometry and mass across materials	Separates universal mass coupling from carrier-dependent occupancy or relaxation
Event-resolved simulation	Integrate q_total(t), actual plane crossings, intra-cell phase, node distance, sheet, polarity, and state	Must recover the exact cycle impulse without inserting the final cross product

Principal findings

Finding	Corrected exact statement
Geometry	The source Cell3, all 27 nodes, and q_i=-1,0,+1 plane registries are rendered with equal physical coordinate scales.
Plane spacing	Adjacent centered half-step planes are separated by one half of the source primitive boundary spacing.
Absolute path	The material path is a translated circle/generalized screw path; the full-scale translation is 443.775 m per revolution versus 4-5 mm local radius.
Actual crossings	The total a, b, and c crossing rates remain one-signed; rotation advances and retards timing.
Exact kernel	v_R^T G_phase V_E = qdot_R^T W qdot_E and d v=-(kappa_F/rho)u[(Wqdot_E)^T d q_R].
Radial law	qdot_R ~ rho and impulse coefficient ~1/rho, producing exact radius cancellation across the wall.
Geometric pump	Zero closed q_R displacement combines with nonzero vector-weighted phase integral to produce Delta v_rev.
Recovered acceleration	0.074911384073992 m/s^2 in the maximum pure-lateral orientation.
Recovered force	0.001694455587915 N for the 22.6195 g tube at 50,000 rpm.
Remaining closure	The source data do not yet select a unique node-, plane-, sheet-, or distributed intra-cell localization mechanism.

Canonical synthesis

The corrected reduction identifies a precise physical target. Earth translation supplies the baseline Cell3 address current, while local rotation supplies a bounded phase modulation of that current. The address coupling W mixes the three rotation coordinates with the three translation coordinates. The vector-weighted closed modulation produces the lateral impulse, and G_phase is the exact physical-space representation of the same ordered address coupling.

The source data and exact algebra determine the coordinate-differential kernel, its radius cancellation, its reversal law, its cycle impulse, and the recovered tube force. They do not yet select a unique microscopic carrier or prove that momentum transfer is localized exclusively at a node, a face boundary, a sheet transition, or a distributed intra-cell process.

Audited invariant	Verified value
Matrix closure max error	1.77635684e-15
All 27 node max source error	0 nm
Mean lateral acceleration	0.074911384073992 m/s^2
Tube force at 50,000 rpm	0.001694455587915 N
Velocity increment per revolution	8.989366088879059e-05 m/s
Actual crossing-rate reversals	0

Earth translation establishes the one-signed Cell3 traversal current; rotation advances and retards its phase; the W-weighted, vector-ordered closed modulation generates the lateral acceleration. The remaining physical task is to identify the material degree of freedom and spatial localization that carry this coordinate-differential momentum transfer.

Appendix A. Complete matrices and vector ledger

Cell3 basis B

Cell3 basis B	1	2	3
1	0.327501242069651	0.035330775844051	-0.010129317626665
2	-0.054897007886313	0.207319850441498	-0.389787702363804
3	-0.037487502896305	-0.547401085391957	-0.689734408362922

Dimension: nm

Direct metric M=B^T B

Direct metric M=B^T B	1	2	3
1	0.111676057905435	0.020710333281606	0.043937235097789
2	0.020710333281606	0.34387773239712	0.296392758961523
3	0.043937235097789	0.296392758961523	0.627770610069385

Dimension: nm^2

Reciprocal metric R=M^(-1)

Reciprocal metric R=M^(-1)	1	2	3
1	9.208025528683773	0.001534802428118	-0.645188035676895
2	0.001534802428118	4.903398312306197	-2.315175584416388
3	-0.645188035676895	-2.315175584416388	2.731171274746109

Dimension: nm^-2

Address coupling W

Address coupling W	1	2	3
1	0.083353302633457	0.137500576603588	0.102559211221958
2	0.137500576603588	0.692400042848548	0.55080090445347
3	0.102559211221958	0.55080090445347	0.587747652353391

Dimension: nm^2

Physical response G_phase

Physical response G_phase	1	2	3
1	2.9764106378	1.2431507795	-2.0803600354
2	1.2431507795	2.5922286635	-2.3178839937
3	-2.0803600354	-2.3178839937	6.7735709133

Dimension: dimensionless

Core vectors

Vector	x	y	z	Dimension
V_E	-3.590031990658130e+05	76679.886034001989174	-44701.060100687995146	m/s
G_phase V_E	-8.802219816350914e+05	-1.439114365615995e+05	2.663350269486193e+05	m/s
n_opt	-0.31586503640732	0.646162683023588	-0.694768354088729	unit
lateral direction	0.077469585421515	0.747375863362418	0.659869519070171	unit
a_bar	0.005803353867564	0.055986960347974	0.049431738981786	m/s^2
Delta v_rev	6.964024641076892e-06	6.718435241756831e-05	5.931808677814329e-05	m/s

Appendix B. All 27 centered Cell3 nodes

Address	lambda	polarity	kind	ICRS x (nm)	ICRS y (nm)	ICRS z (nm)
(-1, -1, -1)	-3	-1	corner	-0.176351350144	0.118682429904	0.637311498326
(-1, -1, 0)	-2	1	edge center	-0.181416008957	-0.076211421278	0.292444294144
(-1, 0, -1)	-2	1	edge center	-0.158685962221	0.222342355125	0.36361095563
(0, -1, -1)	-2	1	edge center	-0.012600729109	0.091233925961	0.618567746877
(-1, -1, 1)	-1	-1	corner	-0.18648066777	-0.271105272459	-0.052422910037
(-1, 0, 0)	-1	-1	face center	-0.163750621035	0.027448503943	0.018743751448
(-1, 1, -1)	-1	-1	corner	-0.141020574299	0.326002280346	0.089910412934
(0, -1, 0)	-1	-1	face center	-0.017665387922	-0.103659925221	0.273700542696
(0, 0, -1)	-1	-1	face center	0.005064658813	0.194893851182	0.344867204181
(1, -1, -1)	-1	-1	corner	0.151149891926	0.063785422018	0.599823995429
(-1, 0, 1)	0	1	edge center	-0.168815279848	-0.167445347239	-0.326123452733
(-1, 1, 0)	0	1	edge center	-0.146085233113	0.131108429164	-0.254956791248
(0, -1, 1)	0	1	edge center	-0.022730046735	-0.298553776403	-0.071166661485
(0, 0, 0)	0	1	body center	0	0	0
(0, 1, -1)	0	1	edge center	0.022730046735	0.298553776403	0.071166661485
(1, -1, 0)	0	1	edge center	0.146085233113	-0.131108429164	0.254956791248
(1, 0, -1)	0	1	edge center	0.168815279848	0.167445347239	0.326123452733
(-1, 1, 1)	1	-1	corner	-0.151149891926	-0.063785422018	-0.599823995429
(0, 0, 1)	1	-1	face center	-0.005064658813	-0.194893851182	-0.344867204181
(0, 1, 0)	1	-1	face center	0.017665387922	0.103659925221	-0.273700542696
(1, -1, 1)	1	-1	corner	0.141020574299	-0.326002280346	-0.089910412934
(1, 0, 0)	1	-1	face center	0.163750621035	-0.027448503943	-0.018743751448
(1, 1, -1)	1	-1	corner	0.18648066777	0.271105272459	0.052422910037
(0, 1, 1)	2	1	edge center	0.012600729109	-0.091233925961	-0.618567746877
(1, 0, 1)	2	1	edge center	0.158685962221	-0.222342355125	-0.36361095563
(1, 1, 0)	2	1	edge center	0.181416008957	0.076211421278	-0.292444294144
(1, 1, 1)	3	-1	corner	0.176351350144	-0.118682429904	-0.637311498326

Appendix C. Complete plane-family and node registry

Channel	Face family	Area nm^2	Primitive spacing nm	Half-step spacing nm	Normal
a	bc	0.357809539128562	0.329546529709293	0.164773264854647	(-0.995965190309, 0.083602162507, -0.032619290564)
b	ac	0.261106235799531	0.45159737972413	0.225798689862065	(0.089052594102, 0.866576769291, -0.491033948326)
c	ab	0.1948691654699	0.605097741514743	0.302548870757372	(0.194092200808, 0.913177168821, 0.358379234791)

a-channel / bc node planes

q level	Nine centered node addresses on the plane
-1	(-1, -1, -1), (-1, -1, 0), (-1, -1, 1), (-1, 0, -1), (-1, 0, 0), (-1, 0, 1), (-1, 1, -1), (-1, 1, 0), (-1, 1, 1)
0	(0, -1, -1), (0, -1, 0), (0, -1, 1), (0, 0, -1), (0, 0, 0), (0, 0, 1), (0, 1, -1), (0, 1, 0), (0, 1, 1)
1	(1, -1, -1), (1, -1, 0), (1, -1, 1), (1, 0, -1), (1, 0, 0), (1, 0, 1), (1, 1, -1), (1, 1, 0), (1, 1, 1)

b-channel / ac node planes

q level	Nine centered node addresses on the plane
-1	(-1, -1, -1), (-1, -1, 0), (-1, -1, 1), (0, -1, -1), (0, -1, 0), (0, -1, 1), (1, -1, -1), (1, -1, 0), (1, -1, 1)
0	(-1, 0, -1), (-1, 0, 0), (-1, 0, 1), (0, 0, -1), (0, 0, 0), (0, 0, 1), (1, 0, -1), (1, 0, 0), (1, 0, 1)
1	(-1, 1, -1), (-1, 1, 0), (-1, 1, 1), (0, 1, -1), (0, 1, 0), (0, 1, 1), (1, 1, -1), (1, 1, 0), (1, 1, 1)

c-channel / ab node planes

q level	Nine centered node addresses on the plane
-1	(-1, -1, -1), (-1, 0, -1), (-1, 1, -1), (0, -1, -1), (0, 0, -1), (0, 1, -1), (1, -1, -1), (1, 0, -1), (1, 1, -1)
0	(-1, -1, 0), (-1, 0, 0), (-1, 1, 0), (0, -1, 0), (0, 0, 0), (0, 1, 0), (1, -1, 0), (1, 0, 0), (1, 1, 0)
1	(-1, -1, 1), (-1, 0, 1), (-1, 1, 1), (0, -1, 1), (0, 0, 1), (0, 1, 1), (1, -1, 1), (1, 0, 1), (1, 1, 1)

Appendix D. Branch ledger

Channel	qdot_E s^-1	qdot_R amp s^-1	rate fraction	q_R amp	baseline/rev	positive variation/rev	max phase shift deg	W qdot_E m^2/s	mean lateral a m/s^2	share
a	-2.2177348429e+15	1.4621742595e+11	6.5930977465e-05	2.7925471327e+07	2.6612818114e+12	5.5850942654e+07	0.0037775667	-0.0001451789	0.0652044635	87.042129%
b	2.4990658631e+14	5.6553110703e+10	0.000226297	1.0800848539e+07	2.9988790357e+11	2.1601697078e+07	0.012965863	-0.0001033634	0.0132142276	17.639812%
c	5.1817517757e+13	8.3075987258e+10	0.0016032414	1.5866344829e+07	6.2181021309e+10	3.1732689658e+07	0.091858963	-5.9344738002e-05	-0.003507307	-4.681941%

Appendix E. Ordered channel-pair ledger

Rot ch.	Earth ch.	W_ij nm^2	a_x	a_y	a_z	lateral comp.	lateral share	nonlateral magnitude
a	a	0.083353302633	0.003341231211	0.061545219431	0.055720508525	0.083024520468	110.830312%	0.003268322542
a	b	0.137500576604	-0.000621092549	-0.011440476518	-0.01035773656	-0.01543320644	-20.601951%	0.000607539751
a	c	0.102559211222	-9.605619395765e-05	-0.001769347633	-0.001601894522	-0.002386850517	-3.186232%	9.396016146688e-05
b	a	0.137500576604	-0.030951763534	0.023581686088	0.036003633903	0.038984263299	52.040506%	0.035924872357
b	b	0.692400042849	0.017563297101	-0.013381213592	-0.020429935059	-0.022121266138	-29.529912%	0.020385242535
b	c	0.550800904453	0.002896960071	-0.002207150585	-0.003369794736	-0.00364876961	-4.870781%	0.003362422973
c	a	0.102559211222	0.052395657941	-0.001309843149	-0.025039033607	-0.01344237032	-17.944363%	0.056509083733
c	b	0.550800904453	-0.031709058946	0.000792697244	0.015153244062	0.008135119008	10.859657%	0.03419844196
c	c	0.587747652353	-0.007015821233	0.000175389064	0.003352746974	0.001799944324	2.402765%	0.007566612294

Appendix F. Radial wall ledger

rho mm	qdot_R amp a	qdot_R amp b	qdot_R amp c	|dv/dq| a	|dv/dq| b	|dv/dq| c	mean a magnitude	error
4	1.1697394076e+11	4.5242488563e+10	6.6460789806e+10	1.1157180134e-12	7.9436052018e-13	4.5607186312e-13	0.074911384074	4.78544781e-17
4.125	1.2062937641e+11	4.6656316330e+10	6.8537689488e+10	1.0819083766e-12	7.7028898926e-13	4.4225150363e-13	0.074911384074	7.38073366e-17
4.25	1.2428481206e+11	4.8070144098e+10	7.0614589169e+10	1.0500875420e-12	7.4763343076e-13	4.2924410646e-13	0.074911384074	5.63985255e-17
4.375	1.2794024771e+11	4.9483971866e+10	7.2691488851e+10	1.0200850408e-12	7.2627247559e-13	4.1697998914e-13	0.074911384074	5.28521592e-17
4.5	1.3159568335e+11	5.0897799633e+10	7.4768388532e+10	9.9174934526e-13	7.0609824016e-13	4.0539721166e-13	0.074911384074	5.02246636e-17
4.625	1.3525111900e+11	5.2311627401e+10	7.6845288213e+10	9.6494530890e-13	6.8701450394e-13	3.9444053026e-13	0.074911384074	5.33268782e-17
4.75	1.3890655465e+11	5.3725455168e+10	7.8922187895e+10	9.3955201130e-13	6.6893517489e-13	3.8406051631e-13	0.074911384074	5.10565987e-17
4.875	1.4256199030e+11	5.5139282936e+10	8.0999087576e+10	9.1546093409e-13	6.5178299092e-13	3.7421281076e-13	0.074911384074	5.06349113e-17
5	1.4621742595e+11	5.6553110703e+10	8.3075987258e+10	8.9257441074e-13	6.3548841614e-13	3.6485749049e-13	0.074911384074	5.28521592e-17

Appendix G. Formula, source, and verification ledger

Object	Formula
Centered half-step position	`x=(1/2)Bq`
Address velocity	`qdot=2B^(-1)v`
Matrix closure	`G_phase=4B^(-T)WB^(-1)`
Mixed scalar	`v_R^T G_phase V_E=qdot_R^T W qdot_E`
Mixed energy per mass	`epsilon_cross=kappa_F qdot_R^T W qdot_E`
Coordinate-differential acceleration	`a=-(kappa_F/rho)u(qdot_R^T W qdot_E)`
Differential impulse	`d v=-(kappa_F/rho)u[(Wqdot_E)^T d q_R]`
Actual trajectory	`q_total(t)=q_0+qdot_E t+q_R(t)`
First-order crossing shift	`delta t_i,k=-q_R,i(t_i,k^0)/qdot_E,i`
Cycle pump	`Delta v_rev=pi kappa_F n x (G_phase V_E)`
Uniform tube force	`F=M(kappa_F omega/2)n x (G_phase V_E)`
Forward/reverse diagnostic	`a_odd=(a_+-a_-)/2; a_even=(a_++a_-)/2`

Source ledger

File	Role	SHA-256
K-Field_quantum_discrete_gravity_states_1784549003(1).json	Physical nanometre Cell3 basis, nodes, planes, W, G_phase, metrics	ba81032bb185d3ff1984e42acefd7270b8424b78eb7f335b01d3aa985283cf1e
K-Field_cell_level_lateral_acceleration_event_reduction.json	Prior exact numerical event reduction used as independent comparison	b55298c36c81a590b13572f69b5dc0c9c1b0db460a24e49eded220bdfca893de
K-Field_dual_axis_kinetic_mass_CMB_traversal_geometry_1784745791.json	Absolute Earth-CMB translation plus local rotational path formulation	bac88436eb33aa988ed66b6f61f2a2896696fc4c4e7603e9b611211306ecb13b
K-Native_Cell3_antipodal_sphere_theorem1784492216(2).json	Antipodal and centered-node structural reference	2c6331de64d582f58b287f8bda04ca2cf90e70e7eae1c1d463b5f1ec8cfa5005
K-Field_Consequences_of_the_Native_Cell3_Antipodal_Sphere_Theorem_1784494389(2).json	Antipodal consequences and projective/cell registry reference	6e7004c122f1c9e96424e7f55b931236e37e08be36ca3239718b11c1d6ffed07
K-Field_single_cell3_lateral_force_reduction_1784915217.pdf	Superseded report under audit	2cb06fce8411dd07c1978598a3c1df6a7bda9c8b493e3453f03b052dc69b8169
K-Field_single_cell3_lateral_force_reduction_1784915217.json	Superseded machine-readable report under audit	854ea3b694e8a26bfcb6fed30f5d2933c9e64bf9121e54547f470d428f135b77

Numerical verification

Check	Value
G_identity_max_abs_error	1.776356839400e-15
direct_metric_max_abs_error	0
reciprocal_metric_max_abs_error	0
volume_error_nm3	0
node_source_max_error_nm	0
face_area_max_error_nm2	0
face_spacing_max_error_nm	5.551115123126e-17
mean_acceleration_source_error_m_s2	0
cycle_endpoint_error_m_s	1.954866896716e-19
pairwise_sum_error_m_s2	2.046109832890e-16
wall_integration_error_m_s2	5.605063087764e-17
total_rate_direction_reversal_count	0

Archival record

Field	Value
Document title	K-Field Single-Cell3 Lateral Force Reduction
Principal investigator	P. Dwayne Esterline
Organization	Krytronx
Report generation	Python ReportLab; built-in core fonts only
Figure generation	Python/Matplotlib from the numerical ledgers; raster images inserted without aspect-ratio distortion
Generation epoch	1784985105
Generated UTC	2026-07-25T13:11:45Z
Output PDF	K-Field_single_cell3_lateral_force_reduction_1784985105.pdf
Output JSON	K-Field_single_cell3_lateral_force_reduction_1784985105.json
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.